Reviewer Pack — Matroid Rank Bounded by Nisan-Wigderson Seed Length for 3-SA...

Entry fa70962e79f5 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Matroid Rank Bounded by Nisan-Wigderson Seed Length for 3-SAT

Entry ID: `fa70962e79f5` **Verdict**: `INCONCLUSIVE` **Recorded**: 2026-05-12 02:14:52 UTC

1. Conjecture

Field A (mathematical branch): Matroid Theory **Field B** (complexity object): Nisan-Wigderson PRG Seed Length

Statement:

> For any 3-SAT instance with n variables, the seed length of the Nisan-Wigderson PRG fooling it is at most the rank of the matroid formed by its clause hypergraph, where the rank is the maximal size of a set of clauses with pairwise disjoint variable support.

Rationale (proposer's reasoning):

> Matroid rank captures combinatorial dependencies between clauses, which may constrain the pseudorandomness needed to fool the formula. This bridges structural properties of CNF formulas with PRG efficiency, leveraging matroid theory's role in encoding combinatorial constraints.

Taxonomy category: `NISAN_WIGDERSON_DESIGNS` (status at proposal time: partially_alive)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): `3c5b26976af504a7`

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

> default: >=80% seeds must support, no counterexample

3. Barrier filter (F1)

Final: `PASS`

Barrier	Verdict	Confidence	LLM_1	LLM_2
RELATIVIZATION	SAFE	0.00	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.00	UNCERTAIN	UNCERTAIN

Barrier	Verdict	Confidence	LLM_1	LLM_2
ALGEBRIZATION	SAFE	0.95	SAFE	SAFE
KARP_LIPTON	SAFE	0.95	SAFE	SAFE

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)
Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution**: rc=124, elapsed=240.0s

5.1 Generated Python source

```python
import random
from itertools import combinations

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def nisan_wigderson_seed_length(n):
        # Placeholder for actual Nisan-Wigderson seed length calculation
        return 2 * n

    def matroid_rank(clauses):
        variables = set()
        for clause in clauses:
            variables.update(clause)

        max_independent_set_size = 0

        for r in range(1, len(variables) + 1):
            for subset in combinations(variables, r):
                independent = True
                for clause in clauses:
                    if not any(var in subset for var in clause):
                        independent = False
                        break
                if independent:
                    max_independent_set_size = max(max_independent_set_size, len(subset))

        return max_independent_set_size

    n = random.randint(5, 40)
    clauses = []
    for _ in range(3 * n):
        clause = [random.choice(range(n)) for _ in range(3)]
        clauses.append(clause)

    rank = matroid_rank(clauses)
    seed_length = nisan_wigderson_seed_length(n)

    return {
```

```
        "metric_name": "seed_length",
        "metric_value": seed_length,
        "instances_tested": 1,
        "conjecture_holds": seed_length <= rank,
        "counterexample": "" if seed_length <= rank else f"Seed {seed} violates the conjecture"
    }

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] or list(range(2, 30))  # Default to first 30 primes if

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_metric_value = sum(r["metric_value"] for r in results) / len(results)
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_metric_value} std=0.0 support_fraction=1.0")
    elif sum(1 for r in results if not r["conjecture_holds"]) / len(results) >= 0.2:
        first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
        print(f"RESULT: FALSIFIED counterexample=\"First failing seed\" first_failing_seed={first_
    else:
        print(f"RESULT: INCONCLUSIVE support_fraction={support_fraction:.2f}")
```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```
TIMEOUT after 240s
```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

> skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

> Test timed out before producing results, preventing evaluation of support fraction or counterexamples. | next: Run test with extended timeout and memory limits to complete execution

11. Audit log (LLM calls)

Total LLM calls: 7

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	qwen3:8b	0	0	107456
2	preregistration	ollama_remote	qwen3:8b	0	0	23912
3	novelty	ollama_remote	qwen3:8b	0	0	22149
4	novelty	ollama_remote	qwen3:8b	0	0	16268
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	13604
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	6855
7	judge	ollama_remote	qwen3:8b	0	0	16193

Totals: 0 input tokens, 0 output tokens, ~$0.0000 cost, 206438 ms total latency. Provider mix: {'ollama_remote': 7}

(full prompt+response transcripts available in `research/audit/fa70962e79f5.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/fa70962e79f5.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)
Replay tarball: `research/replay/fa70962e79f5.tar.gz` (if generated)